

WEB APPLICATION RECONNAISSANCE WITH WHATWEB

A Web Technology Fingerprinting Proof of Concept

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Tool: WhatWeb

Target: Metasploitable 2

Target IP: 192.168.56.101

Environment: Kali Linux and Metasploitable 2 Virtual Machines

1. Executive Summary

This Project demonstrates web application reconnaissance and technology fingerprinting using WhatWeb against a deliberately vulnerable Metasploitable 2 virtual machine in an isolated laboratory environment. The objective was to identify technologies exposed by the target web server, including the web server software, operating system, server-side technologies, versions, HTTP headers, and WebDAV configuration.

The reconnaissance identified Apache HTTP Server 2.2.8 running on Ubuntu Linux, PHP version 5.2.4-2ubuntu5.10, and WebDAV version 2. The web application returned a **200 OK** response and was identified by the page title as **Metasploitable2 - Linux**. WhatWeb did not identify a specific CMS or JavaScript library from the scan performed.

2. Theory and Background

2.1 Web Application Reconnaissance

Web application reconnaissance is the process of gathering information about a web application and its underlying infrastructure before performing further security assessment.

During reconnaissance, a security professional may attempt to identify:

- IP addresses
- Open services
- Web servers
- Operating systems
- Programming languages
- Frameworks
- Content Management Systems (CMS)
- JavaScript libraries
- HTTP headers
- Server versions
- Technologies exposed by the application

The information gathered during reconnaissance helps security professionals understand the attack surface and determine which areas may require further investigation.

2.2 Web Technology Fingerprinting

Web technology fingerprinting is the process of identifying technologies used by a website or web server based on observable characteristics.

These characteristics can include:

- HTTP response headers
- Server banners
- Page titles
- Cookies
- HTML content
- File extensions
- Response behavior
- Technology-specific patterns

For example, an HTTP response containing:

Server: Apache/2.2.8 (Ubuntu)

can reveal that the web server is Apache version 2.2.8 running on Ubuntu Linux.

Similarly:

X-Powered-By: PHP/5.2.4-2ubuntu5.10

reveals that the application is using PHP and exposes the specific PHP version.

2.3 WhatWeb

WhatWeb is a web technology fingerprinting tool used to identify technologies associated with websites and web applications.

WhatWeb can identify technologies such as:

- Web servers
- Operating systems
- Programming languages
- JavaScript frameworks
- CMS platforms
- Web application frameworks
- HTTP headers
- Plugins
- Cookies
- Server versions

WhatWeb is useful during reconnaissance because it allows security analysts to quickly build an initial technology profile of a web application.

3. Lab Environment

The assessment was performed in a controlled virtual laboratory environment.

Component	Details
Attacker/Assessment VM	Kali Linux
Target VM	Metasploitable 2
Reconnaissance Tool	WhatWeb
Virtualization Platform	Oracle VirtualBox
Network Type	Host-Only Network
Target IP	192.168.56.101

Target Web Service	HTTP
Assessment Type	Web application reconnaissance

The Kali Linux virtual machine was configured with access to the Host-Only network so that it could communicate directly with the Metasploitable 2 virtual machine.

Kali Linux also had a NAT adapter available for Internet connectivity and package installation. Metasploitable 2 was kept on the isolated Host-Only network.

4. Lab Network Configuration

The two virtual machines were connected through a VirtualBox Host-Only network.

The Host-Only configuration provided an isolated network between the virtual machines while preventing the intentionally vulnerable Metasploitable 2 system from being directly exposed to the external network.

The resulting target IP address was:

Metasploitable 2: 192.168.56.101

The Kali Linux machine was able to communicate with the target successfully.

5. Objectives

The objectives of this Project were to:

1. Configure an isolated reconnaissance laboratory.
2. Establish network connectivity between Kali Linux and Metasploitable 2.
3. Identify the IP address of the target web server.
4. Verify connectivity to the target.
5. Confirm that the target web service was accessible.
6. Install and run WhatWeb.
7. Identify technologies used by the target web server.
8. Identify web server and PHP versions.
9. Identify HTTP headers and WebDAV information.

10. Analyse the security implications of the technologies identified.

6. Methodology and Step-by-Step Implementation

6.1 Identifying the Target IP Address

The first step was to identify the IP address assigned to the Metasploitable 2 virtual machine.

The following command was used:

```
ifconfig
```

The network interface configuration showed the target IP address:

```
192.168.56.101
```

This IP address was subsequently used as the target for reconnaissance.

Screenshot 2: Metasploitable 2 IP Address

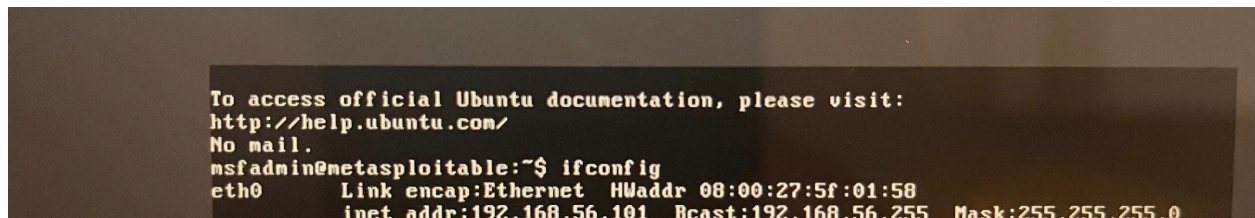


Figure 2: Metasploitable 2 network configuration showing the target IP address 192.168.56.101.

6.2 Testing Network Connectivity

Before performing web reconnaissance, connectivity between Kali Linux and the target was verified.

The following command was used from Kali Linux:

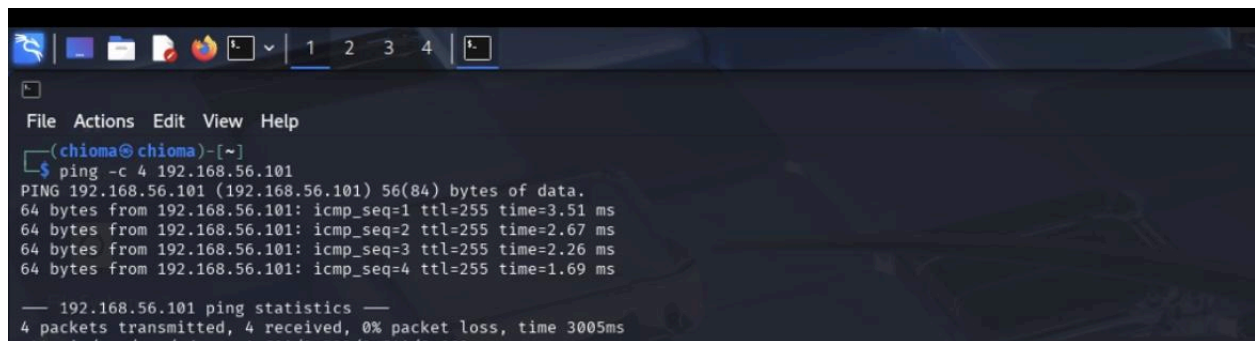
```
ping -c 4 192.168.56.101
```

The **ping** command tests network connectivity between two hosts.

The **-c 4** option instructs Kali Linux to send four ICMP packets to the target.

The successful responses confirmed that Kali Linux could communicate with Metasploitable 2 over the Host-Only network.

Screenshot 3: Successful Ping

A screenshot of a Kali Linux terminal window. The window title is "chioma". The terminal shows the command `ping -c 4 192.168.56.101` being executed. The output shows four successful ping responses from 192.168.56.101 to 192.168.56.101, each with 64 bytes of data and varying times. The ping statistics at the bottom show 4 packets transmitted, 4 received, 0% packet loss, and a time of 3005ms.

```
File Actions Edit View Help
(chioma@chioma)-[~]
$ ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=255 time=3.51 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=255 time=2.67 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=255 time=2.26 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=255 time=1.69 ms

— 192.168.56.101 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
```

Figure 3: Successful ICMP connectivity test between Kali Linux and Metasploitable 2.

6.3 Manual HTTP Header Inspection

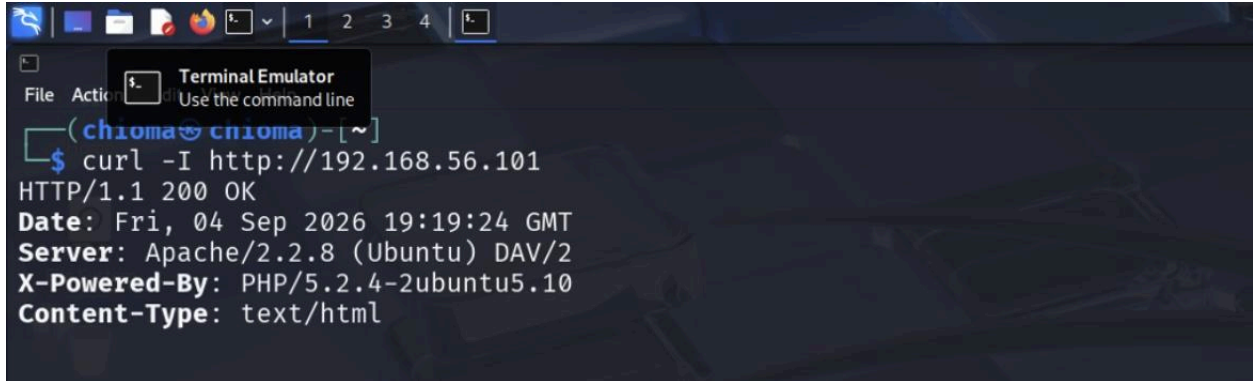
Before performing technology fingerprinting with WhatWeb, the HTTP response headers of the target web server were manually inspected using cURL.

The following command was used:

```
curl -I http://192.168.56.101
```

The **-I** option instructs CURL to retrieve only the HTTP response headers rather than the full webpage content.

The target returned the following response:

A screenshot of a terminal emulator window. The title bar says "Terminal Emulator" and "Use the command line". The prompt is "(chioma@chioma)-[~]". The command entered is "\$ curl -I http://192.168.56.101". The output is: "HTTP/1.1 200 OK", "Date: Fri, 04 Sep 2026 19:19:24 GMT", "Server: Apache/2.2.8 (Ubuntu) DAV/2", "X-Powered-By: PHP/5.2.4-2ubuntu5.10", and "Content-Type: text/html".

```
Terminal Emulator
Use the command line

(chioma@chioma)-[~]
$ curl -I http://192.168.56.101
HTTP/1.1 200 OK
Date: Fri, 04 Sep 2026 19:19:24 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Type: text/html
```

The response confirmed that the web server was accessible and returned an HTTP **200 OK** status. The **Server** header revealed that the target was running Apache HTTP Server version 2.2.8 on Ubuntu Linux, with WebDAV enabled. The **X-Powered-By** header also revealed the use of PHP version 5.2.4-2ubuntu5.10.

This manual inspection provided an initial view of the technologies exposed by the target and was later corroborated by the WhatWeb fingerprinting scan.

6.4 Checking the WhatWeb Installation

The WhatWeb installation was verified using:

```
whatweb --version
```

This confirmed that WhatWeb was installed and available for use.

6.5 Performing Web Technology Fingerprinting

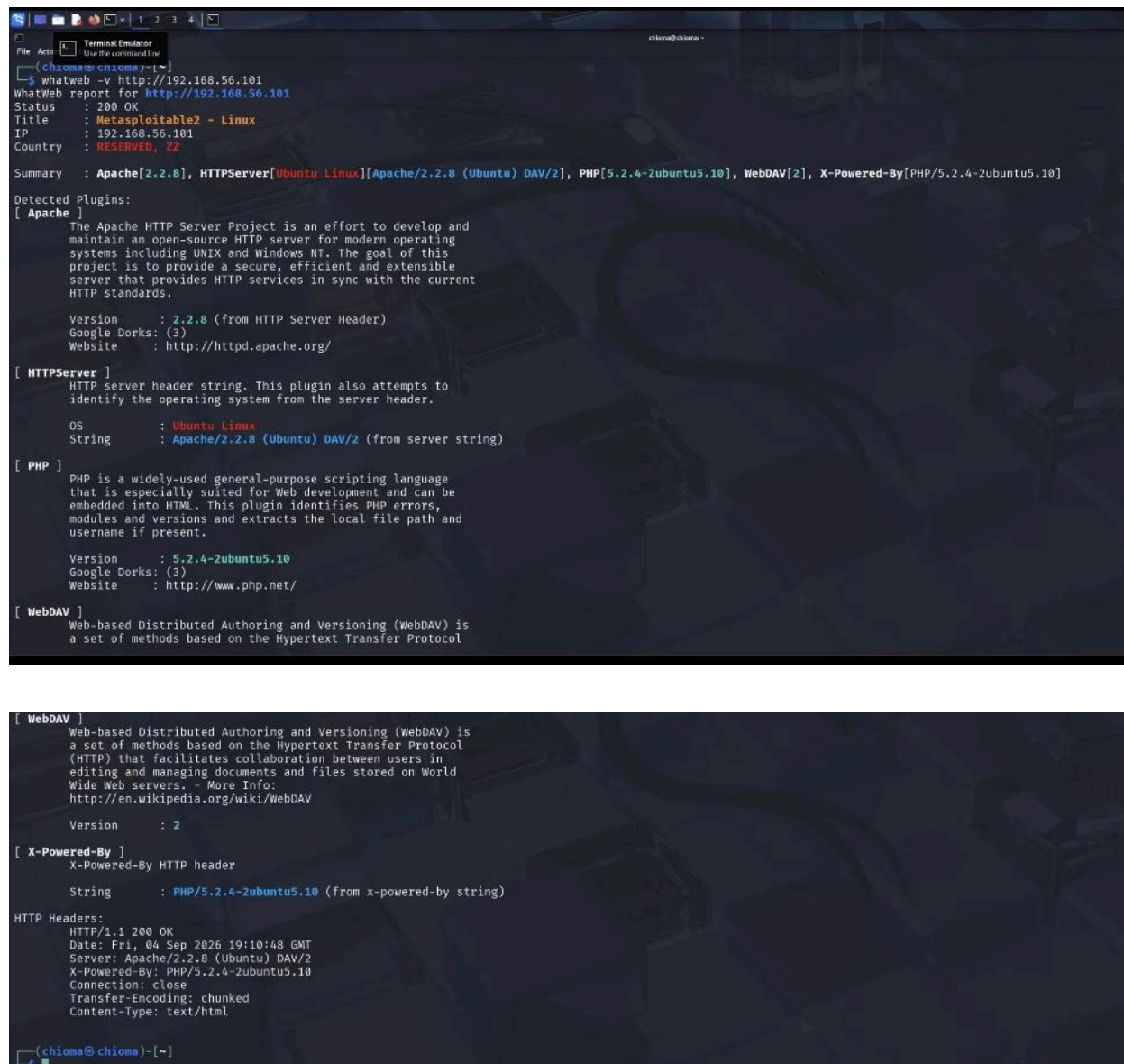
The primary reconnaissance scan was performed using:

```
whatweb -v http://192.168.56.101
```

The **-v** option enables verbose output, providing more information about the technologies detected by WhatWeb.

The scan successfully identified several technologies and configuration details associated with the target web server.

Screenshot 4: WhatWeb Scan



```
(chioma@chioma)~$ whatweb -v http://192.168.56.101
WhatWeb report for http://192.168.56.101
Status      : 200 OK
Title       : Metasploitable2 - Linux
IP          : 192.168.56.101
Country     : RESERVED, ZZ

Summary     : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

Detected Plugins:
[ Apache ]
The Apache HTTP Server Project is an effort to develop and
maintain an open-source HTTP server for modern operating
systems including UNIX and Windows NT. The goal of this
project is to provide a secure, efficient and extensible
server that provides HTTP services in sync with the current
HTTP standards.

Version      : 2.2.8 (from HTTP Server Header)
Google Dorks: (3)
Website      : http://httpd.apache.org/

[ HTTPServer ]
HTTP server header string. This plugin also attempts to
identify the operating system from the server header.

OS           : Ubuntu Linux
String       : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)

[ PHP ]
PHP is a widely-used general-purpose scripting language
that is especially suited for Web development and can be
embedded into HTML. This plugin identifies PHP errors,
modules and versions and extracts the local file path and
username if present.

Version      : 5.2.4-2ubuntu5.10
Google Dorks: (3)
Website      : http://www.php.net/

[ WebDAV ]
Web-based Distributed Authoring and Versioning (WebDAV) is
a set of methods based on the Hypertext Transfer Protocol

Version      : 2

[ X-Powered-By ]
X-Powered-By HTTP header

String       : PHP/5.2.4-2ubuntu5.10 (from x-powered-by string)

HTTP Headers:
HTTP/1.1 200 OK
Date: Fri, 04 Sep 2026 19:10:48 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Connection: close
Transfer-Encoding: chunked
Content-Type: text/html

(chioma@chioma)~$
```

Figure 4: WhatWeb verbose scan against the Metasploitable 2 web server.

6.6 Saving the WhatWeb Results

The scan results can also be saved to a text file for documentation and inclusion in the GitHub repository.

The command used for the final evidence file is:

```
whatweb -v http://192.168.56.101 | tee whatweb-results.txt
```

The `tee` command displays the results in the terminal while simultaneously writing them to `whatweb-results.txt`.

This provides a persistent copy of the reconnaissance output for the project repository.

7. WhatWeb Findings

The WhatWeb scan returned the following results:

Finding	Result
Target	192.168.56.101
HTTP Status	200 OK
Page Title	Metasploitable2 - Linux
Web Server	Apache HTTP Server
Apache Version	2.2.8
Operating System	Ubuntu Linux
Server String	Apache/2.2.8 (Ubuntu) DAV/2
Server-Side Technology	PHP
PHP Version	5.2.4-2ubuntu5.10
WebDAV	WebDAV/2
X-Powered-By	PHP/5.2.4-2ubuntu5.10
Content Type	text/html
Transfer Encoding	chunked
CMS	Not identified by WhatWeb

8. Detailed Findings Analysis

8.1 HTTP Status

The target returned:

HTTP/1.1 200 OK

A **200 OK** response indicates that the HTTP request was successfully processed and that the web server returned the requested content.

This confirms that the web application was accessible from the Kali Linux assessment machine.

8.2 Web Application Title

WhatWeb identified the page title as:

Metasploitable2 - Linux

This provides information about the target application and also confirms that the scanned service corresponds to the Metasploitable 2 system.

Page titles can sometimes provide useful information during reconnaissance because they may disclose application names, product names, administrative interfaces, or other identifying information.

8.3 Apache Web Server

WhatWeb identified:

Apache/2.2.8 (Ubuntu) DAV/2

This indicates that the target is running Apache HTTP Server version 2.2.8 on Ubuntu Linux.

The server software and version are important reconnaissance findings because they provide information about the technology stack that supports the web application.

The version information can be used by security professionals during subsequent vulnerability assessment and patch management activities.

However, identifying an old software version does not by itself prove that the system is vulnerable. Further validation would be required before making a vulnerability determination.

8.4 Operating System

WhatWeb identified:

Ubuntu Linux

This provides an indication of the operating system supporting the web server.

Operating system identification can help security analysts understand the underlying infrastructure and determine which operating-system-specific security controls or vulnerabilities may be relevant during further assessment.

8.5 PHP

The scan identified PHP as a server-side technology.

The detected version was:

PHP/5.2.4-2ubuntu5.10

The version was also exposed through the HTTP response header:

X-Powered-By: PHP/5.2.4-2ubuntu5.10

This is an example of technology and version disclosure.

Exposing server-side technology information can make it easier for an attacker to profile the application and identify potentially relevant vulnerabilities.

8.6 WebDAV

The server response contained:

DAV/2

and WhatWeb identified WebDAV.

WebDAV is an extension to HTTP that provides functionality for managing resources on a web server.

The presence of WebDAV is important from a security perspective because unnecessary or improperly configured WebDAV functionality can increase the attack surface of a web server.

A security assessment should therefore determine whether WebDAV is required and whether it has been securely configured.

8.7 HTTP Headers

The scan returned the following HTTP headers:

HTTP/1.1 200 OK
Date: Fri, 04 Sep 2026 19:10:48 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Connection: close
Transfer-Encoding: chunked
Content-Type: text/html

Several of these headers provide useful reconnaissance information.

The **Server** header exposes:

Apache/2.2.8 (Ubuntu) DAV/2

The **X-Powered-By** header exposes:

PHP/5.2.4-2ubuntu5.10

These headers allow an external observer to identify technologies and versions without requiring authentication.

8.8 CMS Detection

WhatWeb did not identify a specific Content Management System from the scan.

This should be interpreted carefully.

The result does **not** necessarily mean that no CMS exists on the target. It means that the WhatWeb scan performed did not identify a CMS.

Additional enumeration or manual inspection would be required to determine whether a CMS or custom web application is present.

8.9 JavaScript Library Detection

WhatWeb did not identify a JavaScript library in the scan output.

As with CMS detection, this does not prove that the application contains no JavaScript libraries.

It only indicates that no JavaScript library was identified by the WhatWeb scan performed.

9. Security Observations

The reconnaissance exercise identified several pieces of information that would be useful to a security analyst.

9.1 Technology Version Disclosure

The server exposes both Apache and PHP version information through HTTP responses.

For example:

Server: Apache/2.2.8 (Ubuntu) DAV/2

and:

X-Powered-By: PHP/5.2.4-2ubuntu5.10

This information can assist attackers in fingerprinting the target environment.

9.2 Legacy Software

The identified Apache and PHP versions are legacy versions.

Legacy software can present increased security risk because older software may no longer receive security updates or may contain vulnerabilities that have been addressed in newer releases.

The exact security impact should be determined through vulnerability assessment and confirmation of the software's actual patch state.

9.3 WebDAV Exposure

WebDAV functionality was identified on the server.

If WebDAV is not required for business operations, disabling it would reduce the exposed attack surface.

If it is required, its configuration, authentication controls, permissions, and access restrictions should be reviewed.

9.4 Information Disclosure

The HTTP headers provide information about the server's underlying technology stack.

This demonstrates how reconnaissance tools can collect useful information without requiring privileged access to the target.

10. Recommendations

Based on the reconnaissance findings, the following security recommendations are applicable:

10.1 Maintain Updated Software

Web servers and server-side technologies should be maintained on supported versions with current security updates.

Apache and PHP versions should be reviewed against the organization's approved software baseline.

10.2 Review Technology Disclosure

Where appropriate, unnecessary technology and version information should be minimized in HTTP responses.

For example, server configuration can be reviewed to reduce unnecessary exposure of detailed version information.

10.3 Review WebDAV Configuration

Determine whether WebDAV is required.

If it is not required, it should be disabled.

If it is required, access controls, authentication, authorization, and permissions should be reviewed.

10.4 Maintain an Application Technology Inventory

Organizations should maintain an inventory of web applications and their underlying technologies.

This should include:

- Web server
- Operating system
- Programming language
- Frameworks
- CMS
- Third-party libraries
- Software versions

This improves vulnerability management and patching processes.

10.5 Conduct Regular Security Assessments

Web applications should undergo periodic vulnerability assessments and security testing.

Technology fingerprinting can serve as an initial reconnaissance step before deeper security testing.

11. Limitations

This Proof of Concept was limited to web technology fingerprinting using WhatWeb.

The exercise did not attempt to:

- Exploit identified technologies
- Perform password attacks
- Exploit WebDAV
- Conduct vulnerability exploitation
- Modify files on the target
- Perform denial-of-service testing
- Conduct a full penetration test
- Determine whether every detected software version was vulnerable

The absence of a CMS or JavaScript library in the WhatWeb output should also not be interpreted as proof that those technologies do not exist.

WhatWeb provides fingerprinting results based on observable characteristics, and additional techniques may identify technologies that were not detected during this scan.

12. Evidence

The following evidence was collected during the exercise:

1. VirtualBox network configuration.
2. Metasploitable 2 IP address configuration.
3. Successful network connectivity test.
4. WhatWeb verbose scan.
5. HTTP server and PHP version information returned by the target.

Screenshot 5: HTTP Headers and Technology Information

[INSERT SCREENSHOT 5 HERE]

Figure 5: WhatWeb output showing Apache, Ubuntu, PHP, WebDAV, and HTTP header information.

13. Key Results

The reconnaissance successfully identified the following technology stack:

Target: 192.168.56.101

Web Server:
Apache/2.2.8

Operating System:
Ubuntu Linux

Server-Side Technology:
PHP/5.2.4-2ubuntu5.10

WebDAV:
WebDAV/2

HTTP Status:
200 OK

Page Title:
Metasploitable2 - Linux

The assessment demonstrated that WhatWeb can quickly fingerprint technologies exposed by a web application and provide useful information for subsequent security assessment activities.

14. Tools Used

Tool	Purpose
Kali Linux	Security assessment platform
Metasploitable 2	Deliberately vulnerable target
Oracle VirtualBox	Virtualization
WhatWeb	Web technology fingerprinting
cURL	HTTP service verification
Ping	Network connectivity testing

15. Conclusion

This Proof of Concept demonstrated the use of WhatWeb for web application reconnaissance and technology fingerprinting in an isolated laboratory environment.

The assessment successfully identified the target web server as Apache 2.2.8 running on Ubuntu Linux, with PHP version 5.2.4-2ubuntu5.10 and WebDAV version 2 enabled. The HTTP response also exposed server and PHP version information through the **Server** and **X-Powered-By** headers.

The exercise demonstrates how relatively simple reconnaissance can reveal useful information about a web application's underlying technology stack. Such information can support asset identification, vulnerability management, security assessments, and defensive monitoring.

No exploitation was performed. The activity was limited to reconnaissance and technology identification against the intentionally vulnerable Metasploitable 2 machine in an isolated virtual laboratory.

16. Disclaimer

This Proof of Concept was conducted solely in an authorized, isolated laboratory environment using Metasploitable 2, a deliberately vulnerable virtual machine designed for security training and testing.

The techniques demonstrated in this report should only be used against systems for which explicit authorization has been obtained.